
Algorithm 1: Cascaded Assignment

Input: Tracks T ,
Detection partitions $\{D_1, D_2, \dots, D_k\}$, where $D_i \cap D_j = \emptyset$,
Matching thresholds $\{m_1, m_2, \dots, m_k\}$
Output: Matched tracks T_M , unmatched tracks T_U ,
Matched detections D_M , unmatched detections D_U
// Initialize matched and unmatched tracks and detections
 $T_M \leftarrow \emptyset$;
 $D_M \leftarrow \emptyset$;
// All tracks are initially unmatched
 $T_U \leftarrow T$;
 $D_U \leftarrow \emptyset$;
for $i \leftarrow 1$ **to** k **do**
 // Perform assignment between unmatched tracks and current detection
 partition D_i with match threshold m_i
 $(T_{M_i}, T_{U_i}, D_{M_i}, D_{U_i}) \leftarrow \text{Assignment}(T_U, D_i, m_i)$
 // Update the global matched and unmatched sets
 $T_M \leftarrow T_M \cup T_{M_i}$;
 $D_M \leftarrow D_M \cup D_{M_i}$;
 $T_U \leftarrow T_{U_i}$;
 $D_U \leftarrow D_U \cup D_{U_i}$;
end
return T_M, D_M, T_U, D_U
